

LATHIKA M

AI / ML Engineer · Data Scientist · Final Year 2026

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PROFESSIONAL SUMMARY

Final-year AI/ML undergraduate (CGPA 8.44/10) with 3 internships spanning data automation, prompt engineering, and web development. Proven ability to clean large datasets, build and evaluate ML models, and optimize LLM prompts for production systems. Built a real-time GPU-accelerated traffic violation system and a stock prediction model (R^2 : 0.90). Seeking a data science or AI engineering role where I can apply end-to-end ML skills to solve business problems.

EDUCATION

B.E – Computer Science & Engineering (AI & ML)

2022 – 2026

S.A Engineering College, Chennai | CGPA: 8.44 / 10

WORK EXPERIENCE

Prompt Engineering Intern · Lendkraft.ai, Chennai

2025 · 4 Weeks

- Designed and iteratively optimized prompt templates for LLMs used in financial workflows, improving response relevance by 25% measured via human evaluation.
- Tested 50+ prompt variations across zero-shot, few-shot, and chain-of-thought strategies to benchmark output consistency and latency.
- Integrated finalized prompt templates into production AI pipelines, collaborating with the engineering team through agile sprint cycles.
- Documented prompt design patterns and evaluation rubrics, reducing onboarding time for future prompt engineers.

Data Automation Intern · MSME, Chennai

2025 · 6 Weeks

- Processed and cleaned datasets of 10,000+ records using Python and Pandas, improving data quality for downstream reporting by eliminating 15% erroneous entries.
- Built automated ETL pipelines for data cleaning, aggregation, and report generation — reducing weekly manual effort by 40%.
- Conducted EDA using Matplotlib and Seaborn to surface trends and anomalies, which informed 3 actionable business decisions presented to management.

Web Development Intern · 8 Queens, Chennai

2024 · 2 Weeks

- Built responsive frontend components using HTML, CSS, and JavaScript across 3 web pages, improving mobile layout consistency.
- Worked in a 4-member agile team, completing assigned tickets within sprint deadlines.

PROJECTS

Sentiment-Driven LLM Prompt Optimizer

GitHub →

Python · HuggingFace Transformers · NLTK · Pandas · Streamlit

- Built an NLP pipeline that classifies customer review sentiment (positive/negative/neutral) using a fine-tuned BERT model, achieving 88% accuracy on a 5,000-sample test set.
- Created a Streamlit dashboard allowing business users to explore sentiment trends across product categories with interactive charts.
- Applied prompt engineering techniques to generate GPT-style summaries of negative reviews, helping flag critical issues 3× faster than manual review.

Stock Price Prediction using Regression & Ensemble Models

GitHub →

Python · Pandas · NumPy · Scikit-learn · Matplotlib · Seaborn

- Built a multi-model forecasting system comparing Linear Regression, Random Forest, and Gradient Boosting on 5 years of historical stock data.

- Achieved R^2 score of 0.90 with Linear Regression as baseline; Random Forest improved RMSE by 18% after hyperparameter tuning.
- Preprocessed multi-year time-series data including normalization, lag feature engineering, and rolling window statistics.
- Visualized prediction vs actual price curves and feature importance rankings using Matplotlib and Seaborn.

SpeedCamera — Real-Time Traffic Violation Detection System

GitHub →

Python · YOLOv8 · OpenCV · FastAPI · React.js · TensorRT · CUDA

- Built a real-time traffic monitoring pipeline using YOLOv8 for vehicle detection and ByteTrack/Supervision for multi-object tracking across live CCTV feeds.
- Achieved 2× inference speedup using TensorRT + CUDA optimization, enabling 30+ FPS processing on a single GPU.
- Developed FastAPI backend and React.js dashboard to visualize live traffic data and auto-generate violation records with timestamp, speed estimate, and vehicle ID.

TECHNICAL SKILLS

Programming: Python, SQL, HTML/CSS/JavaScript

Machine Learning: Logistic Regression, Random Forest, Gradient Boosting, Neural Networks, BERT

NLP & LLMs: HuggingFace Transformers, Prompt Engineering (zero-shot, few-shot, CoT), NLTK

Computer Vision: YOLOv8, OpenCV, Object Detection, OCR

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, EDA, Statistical Analysis

Tools & Platforms: Scikit-learn, Streamlit, FastAPI, Git, GitHub, Jupyter Notebook

CERTIFICATIONS & ACHIEVEMENTS

- GenAI Powered Data Analytics Job Simulation – Tata (Forage): Practical experience in AI-based risk profiling, data analysis, and business storytelling.
- Completed 3 industry internships across AI, data, and web domains during undergraduate study (2024).
- CGPA 8.44/10 — consistently among top performers in the AI & ML specialization cohort.